THE LCS NETWORK

A Guide to Reticulum, LXMF and LXST

Liberty Communication Systems, Inc. · *lcs.network*

This document explains what the LCS Network is, how it works, and what it requires. It is written for someone who has never encountered Reticulum before. All information applies directly to the LCS products and applications: **Liberty Chat**, **LCS MeshChat**, and the **Liberty Deck**.

Contents

1. The Three-Layer Stack

The LCS Network runs three open protocols, each handling one job.

Layer	Protocol	Function	Analogy
Bottom	Reticulum (RNS)	Addressing, routing, and encryption across any physical medium.	The road network
Middle	LXMF	Text, files, images, and recorded voice — with store-and-forward delivery for offline recipients.	The postal service
Top	LXST	Live, encrypted voice calls.	The telephone

- Reticulum is the foundation. LXMF and LXST ride on top of it and inherit its encryption and routing automatically.
- All three LCS apps — Liberty Chat, LCS MeshChat, and the Liberty Deck — are **LXMF clients**. They speak the same protocol and interoperate fully, including with third-party clients such as Sideband and Nomad Network.

2. Reticulum — The Network Foundation

Reticulum is a cryptography-based networking stack built for links that are slow, intermittent, and untrusted. It does not assume the existence of servers, providers, or fixed infrastructure. It was built by Mark Qvist who spent years perfecting this revolutionary communications breakthrough.

Key characteristics

- **No central authority.** There is no registry, no account server, and no gatekeeper. Any device that runs Reticulum is part of the network.
- **No addresses or ports.** Reticulum uses a single concept — a **destination** — derived from cryptographic keys. There is no DHCP, no subnetting, and no port forwarding.
- **Medium-agnostic.** The same packets travel over LoRa, packet radio, Wi-Fi, Ethernet, USB, Bluetooth, or the internet. These can be mixed freely within a single network.
- **Encrypted by default.** Encryption is structural, not optional. Unencrypted traffic cannot be forwarded across multiple hops.
- **Extreme bandwidth tolerance.** Reticulum remains functional on links as slow as 5 bits per second.
- **Location-independent addresses.** A device keeps the same address whether it is connected over home Wi-Fi, a LoRa radio, or the public internet.

Host device requirement

Reticulum is software. It must run on a **host device** — an Android phone, a Windows/Mac/Linux computer, or a Command Center PRO Gateway. The host provides processing, storage, and the user interface. Physical access to the network is provided by an **interface** attached to or configured on that host. There are various applications which run on Reticulum.

> An RNode radio without a host is just a radio. A host without an interface cannot reach anyone. Together they form a node on the network.

3. Applications and Host Devices

These are applications which run Reticulum on their respective host devices:

- **Liberty Chat** — Android phones and tablets
- **LCS MeshChat** — Windows, macOS, and Linux computers
- **Liberty Deck** — a self-contained handheld with Reticulum, the radio, screen, and keyboard integrated into a single unit
- **Command Center PRO Gateway** — a host device which runs Reticulum continuously as fixed or mobile infrastructure, bridging multiple interface types simultaneously. It routes traffic between LoRa, packet radio, Wi-Fi, and TCP/IP. Hosts a local application of LCS MeshChat accessible by browser at liberty.local.

All applications implement LXMF and interoperate fully. A message composed in Liberty Chat is delivered to LCS MeshChat, the Liberty Deck, or any third-party LXMF client — such as Sideband or Nomad Network — without conversion or additional configuration.

	Liberty Chat	LCS MeshChat	Liberty Deck
Platform	Android phones and tablets	Windows, macOS, Linux	Dedicated LCS handheld
Interface	Native Android app	Desktop app or browser at localhost:8000	On-device — no phone or PC needed
Connections	Bluetooth LE, Wi-Fi, USB-C RNode, TCP	Wi-Fi/Ethernet, USB or IP RNode, TCP, I2P	Built-in LoRa radio; Wi-Fi where supported
Text and files	Yes	Yes (10 MB receive limit)	Yes (device storage limited)
PTT voice	Yes — Codec2 and Opus	Yes — Codec2 and Opus	Limited
Live voice (LXST)	Yes, mid-call codec switching	Yes, mid-call codec switching	Basic
Propagation node	Uses them	Uses and can host one	Uses them
Notable extras	Maps, location sharing, QR identity, multiple identities, Nomad Network	Announce history, peer list, Nomad Network, local message database	Fully standalone — no other device required

- **Liberty Chat** ships with LCS radio defaults: slot 51, 914.875 MHz, 22 dBm, Long Fast preset. The LCS server is listed automatically when a TCP connection is added.
- **LCS MeshChat** is the recommended choice for a machine that remains on continuously, as it can host a propagation node for the whole local segment.
- The **Liberty Deck** carries its own identity, radio, screen, and keyboard. It operates with no other device attached.

Go to start.lcs.network to download Liberty Chat or LCS MeshChat and start chatting.

4. Instances, Nodes and Interfaces — Three Things to Distinguish

These three terms appear throughout the Reticulum documentation and LCS materials. Understanding what each one means prevents a lot of confusion.

Instance

Every host device running Reticulum is called an **instance**. A phone, a laptop, a Command Center Gateway, a solar-powered relay — all are instances. The word simply means "Reticulum is running here."

Node

A node is an instance that has been given a specific role in the network:

Node type	What it does	Enabled by
Client node	An endpoint. Sends and receives traffic for a user. Does not relay for others.	Default — all instances start as clients.
Transport node	A relay. Forwards other devices' encrypted traffic one hop closer to its destination. Cannot read what it carries.	Setting enable_transport = Yes in the configuration.
Propagation node	A store-and-forward mailbox. Holds encrypted messages for offline recipients and delivers them on reconnection.	Running the LXMF propagation node service (built into MeshChat and the Command Center).

A single device can hold more than one role. The Command Center PRO Gateway is simultaneously a transport node, a propagation node, and a protocol bridge.

Interface

An interface is the connection between the Reticulum host and the physical medium. It is not a node — it is the on-ramp a node uses to reach the network. A single host can run several interfaces at once, and Reticulum selects among them automatically based on destination.

Interface	Medium	Use on the LCS Network	Notes
RNode (LoRa)	902–928 MHz ISM	Client radios, solar transport nodes, and IP RNodes. Approx. 341 bps – 21.88 kbps.	Licence-free. Connects by USB-C, Bluetooth LE, or over Wi-Fi (IP RNode).
Packet Radio	VHF 150–174 MHz UHF 450–470 MHz	Long-haul links between Command Center Gateways.	Software modem and PTT interface built into the Command Center. FCC licence required for encrypted operation.
AutoInterface	Local Wi-Fi or Ethernet	Phones and computers joining a Command Center Gateway, or devices sharing one LAN.	Zero configuration. Discovers peers on the same network automatically. Requires UDP ports 29716 and 42671.
TCP Client/Server	Internet or private IP	Connecting to public.lcs.network or a private LCS server.	Client mode requires no public IP or port forwarding.

Interface	Medium	Use on the LCS Network	Notes
I2P	Anonymizing overlay	Reachability without a public IP; endpoint IP addresses are not exposed.	Requires i2pd on the host. Provides a persistent, portable address.
Bluetooth LE	Short-range personal radio	Liberty Chat communicating with a nearby RNode or Android device.	Enabled by default in Liberty Chat.
Serial / USB-C	Direct cable	Client RNode wired to a phone or computer.	Most straightforward client connection.

> - A fresh Liberty Chat install uses AutoInterface and Bluetooth LE only. No traffic reaches the internet until a TCP interface is added or an RNode is attached.
> - The **packet radio software modem** in the Command Center is an interface connected via TCP Server — it drives a push-to-talk radio over VHF or UHF.

Combined Function Hardware

The Solar Transport Node and the IP RNode integrate Reticulum and the radio into one enclosure. They require no separate host and relay traffic automatically, operating like a repeater.

- The **Solar Transport Node** is self-powered for unattended outdoor use.
- The **IP RNode** joins an existing Wi-Fi network, so the radio can sit near the antenna while the user's computer operates anywhere on the LAN.

5. Identities and Addresses

Reticulum Identity

A Reticulum Identity is a public/private key pair generated locally on the host device at first launch. No registration, phone number, or email address is required.

- The **private key** stays on the device. It decrypts inbound messages and produces signed delivery proofs.
- The **public key** is shared with the network so others can encrypt traffic that only the holder can open.
- **Identities are portable files.** Exporting and importing one moves the user's cryptographic identity — and therefore their address — to new hardware.
- A single identity can generate multiple addresses for different purposes: messaging, voice, file hosting.

> Back up the identity file and treat it as carefully as a private key. Whoever holds it controls that address.

LXMF Address

An LXMF address is derived from a Reticulum Identity. It is the address other users send messages to.

- Displayed as a 32-character hexadecimal hash, for example:
  `13425ec15b621c1d928589718000d814`
- The hash is computed from the app name, destination type, and public key, so two users running the same app have distinct, unforgeable addresses.
- No registrar issues it. No authority can revoke it. It cannot be impersonated.

The distinction

The **Reticulum Identity** is the key pair — the master credential. The **LXMF address** is one of the destinations derived from it. Losing the identity loses the address; keeping the identity file backed up means the address survives hardware changes.

6. The Announce System — How Peers Are Discovered

Reticulum has no directory server. Peer discovery is handled by **announces** — small, signed broadcasts that propagate through the network.

What an announce contains

- The destination address (hash)
- The public key for that destination
- Optional application data — in messaging apps, this is the display name
- A unique random value so no two announces are identical
- An Ed25519 signature proving authenticity

How announces propagate

- Any transport node that receives an announce records the direction it came from and re-broadcasts it after a short random delay.
- No node learns the full network topology — each one only knows the **next hop** toward a given destination.
- Announces stop propagating after 128 hops. Announce traffic is capped at 2% of interface bandwidth by default, with closer destinations prioritized, preventing a fast link from flooding a slow LoRa segment.

Practical consequences

- A device that has never announced is not reachable. Press **Announce** once at setup.
- When a device announces, its display name appears in the peer lists of everyone on the network who can receive the announce.
- Re-announce after any change in connection method — switching interfaces, attaching or removing an RNode, or joining a different network.
- Announce sparingly. Flooding announces causes the network to down-prioritize them, making the device **harder** to find.

7. Packets and Links — Two Ways to Move Data

Reticulum can deliver data in two modes. Applications select the appropriate mode automatically.

	Single Packet	Established Link
Best for	Short messages, fire-and-forget	Conversations, voice calls, file transfers
Setup overhead	None	3 packets, 297 bytes total
Encryption	Fresh key derived per packet	Ephemeral shared key for the session
Forward secrecy	With key ratchets enabled	Always
Delivery proof	Optional signed receipt	Built in

	Single Packet	Established Link
Large data	Not supported	Yes — auto-split, compressed, verified, reassembled
Idle cost	None	~0.45 bits per second
Initiator	Remains anonymous	Remains anonymous unless they choose to identify

- A link is not a physical connection. It is an encrypted tunnel that can span many hops and remain open for hours.
- Both ends authenticate cryptographically before any data flows.
- A 1200 bps packet radio channel can sustain approximately 100 concurrent links while retaining most of its capacity for data.

8. LXMF — Messaging

LXMF (Lightweight Extensible Message Format) is the messaging protocol used by all LCS applications. It defines the message structure, addressing, and delivery mechanism.

Message properties

- Protocol overhead is 111 bytes per message — small enough to transit LoRa and packet radio comfortably.
- Every message is timestamped and Ed25519-signed. The sender cannot be forged.
- Messages carry a title, a body, and an extensible **fields** section for attachments, images, audio clips, location data, and telemetry.
- All LXMF clients interoperate: Liberty Chat, LCS MeshChat, the Liberty Deck, Sideband, and Nomad Network exchange messages freely.

Propagation nodes — offline delivery

- A **propagation node** is a store-and-forward message store. When a recipient is offline, the message is held and delivered automatically when they reconnect.
- Propagation nodes peer with each other and synchronize automatically, forming a distributed message store with no single point of failure.
- On the LCS Network, propagation nodes are run by the Command Center PRO Gateway, the LCS server, and optionally by LCS MeshChat on a desktop machine.

> Propagation nodes store messages they cannot read. Content remains encrypted to the recipient's key for as long as it is held.

9. LXST — Real-Time Voice

LXST (Lightweight Extensible Signal Transport) carries live audio between LCS applications, end-to-end encrypted with forward secrecy.

Codecs

Codec	Bit rate range	Suited to
Codec2	700 bps – 3,200 bps	LoRa and packet radio links — intelligible speech at minimal bandwidth
Opus	~4.5 kbps – 96 kbps	Wi-Fi, Ethernet, and internet — natural, high-quality voice

- Codecs can be switched mid-call without dropping the call. Liberty Chat probes the link before dialling and signals any codec change to the remote end, including MeshChat and Sideband peers.

Voice over LoRa

- Live calls require a fast LoRa preset. **Short Fast** (≈10.9 kbps) is the slowest preset that can carry a live call.
- **Half-duplex call** mode is recommended for low link speed.
- **Long Fast** (≈1.07 kbps) is the recommended general-purpose preset for text and PTT, but cannot sustain a live call.
- Where link capacity is insufficient for a live call, you may also use **push-to-talk (PTT) voice messages** instead: recorded clips sent as LXMF messages, delivered on any link speed.
- Liberty Chat PTT has two modes: Low bandwidth (Codec2 1200, about 150 bytes/sec — for LoRa) and High bandwidth (Opus, about 3 KB/sec — for Wi-Fi and TCP).

10. Transport Modes

Each interface on a transport node or gateway can be assigned a **mode** that governs how Reticulum treats the segment behind it. Client devices rarely need to change these; gateways depend on them for correct behaviour.

Mode	Behaviour	Typical use
full	Default. All discovery, meshing, and transport active.	Client devices and most standard links.
gateway	Also resolves unknown paths on behalf of devices behind it.	The LAN-facing interface of a Command Center Gateway.
access_point	Quiet until used — no automatic announces, short-lived paths.	Wide-area radio interfaces where users appear and leave.
roaming	Marks a physically mobile interface.	LoRa side of a vehicle-mounted or portable gateway.
boundary	Marks a transition to a very different network segment.	Internet-facing interface of a gateway whose local side is LoRa.
internal	The slower or private side behind a boundary — shields it from announce floods from larger networks.	Local devices that should not receive the full internet announce stream.

- Setting **boundary** on an internet-facing interface prevents a large internet-connected network from flooding a slow LoRa segment with announces.
- Transport is disabled by default. Client applications do not relay other users' traffic.

11. Security

Cryptographic primitives

These are the authoritative primitives for Reticulum. LXMF and LXST implement no separate cryptography — both inherit these. [1]

Primitive	Purpose
X25519 (ECDH on Curve25519)	Key exchange for establishing shared secrets
Ed25519	Signatures — announces, messages, delivery proofs
AES-256 in CBC mode with PKCS7	Payload encryption
HMAC-SHA256	Message authentication and integrity
HKDF	Key derivation
SHA-256 / SHA-512	Hashing

Properties

- **End-to-end encryption.** Only the destination's private key can decrypt traffic. Relays, gateways, and propagation nodes cannot read what they carry.
- **Forward secrecy.** Links use X25519 to negotiate ephemeral session keys. Compromising a key after the fact does not expose past traffic.
- **Authenticity.** Every announce and message is Ed25519-signed. Addresses cannot be spoofed.
- **Initiator anonymity.** The device that opens a link stays anonymous unless it chooses to identify, and that identification happens inside the encrypted channel.
- **No metadata collection.** No sign-up, no contact list upload, no central log of communications.
- **Optional network isolation.** Interfaces can be assigned a network name and passphrase. Packets without the correct Interface Access Code are discarded at the perimeter.

Limitations

- Encryption protects content, not the existence of a transmission. A radio signal is observable.
- An unlocked device or an exported identity file is equivalent to holding the address.
- Encrypted operation on business-band frequencies requires an FCC licence. Rules vary by jurisdiction.

12. Use-Case Scenarios

Scenario 1 — Two client RNodes with a solar transport node on the mountain

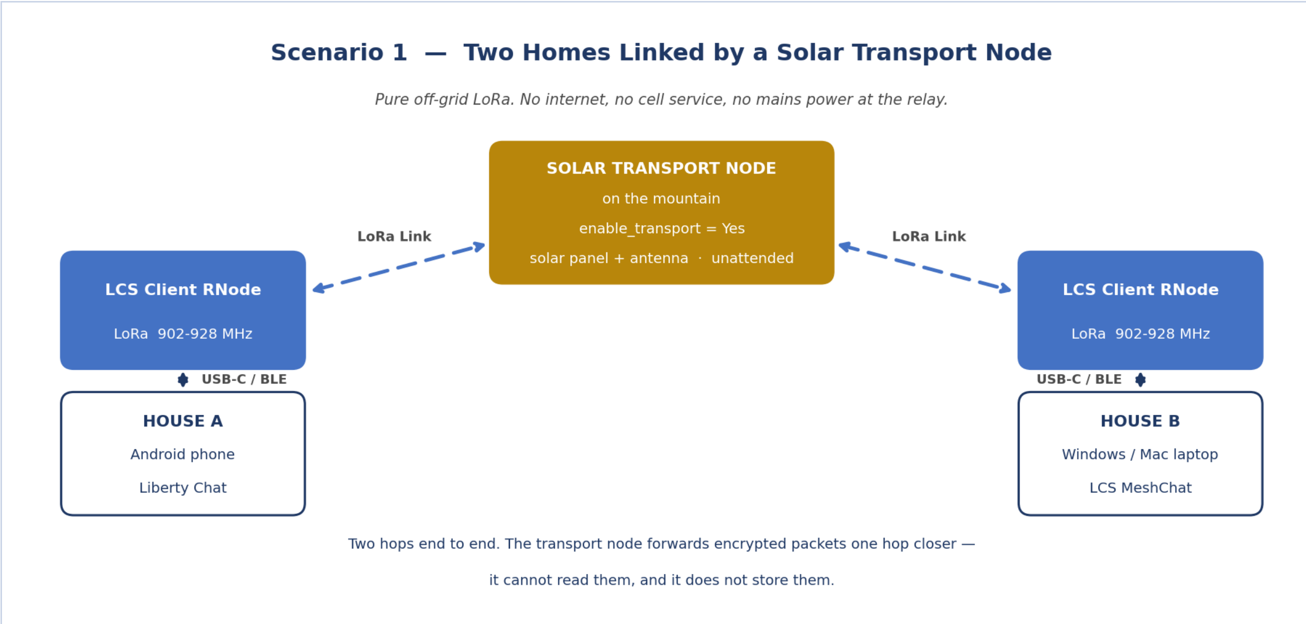

Figure 1 — Off-grid LoRa messaging across a relay.

Requirements

- One LCS Client RNode per household, connected by USB-C or Bluetooth to a phone or computer
- One LCS Solar Transport Node, positioned high with line of sight to both homes
- Matching frequency and preset on every radio unit

How it works

- The transport node re-broadcasts announces and forwards encrypted packets. No other device in the chain needs to be configured to account for it.
- Each user announces once. Both households appear in each other's peer list by display name.
- Messages route automatically across two hops, end-to-end encrypted, with signed delivery receipts.

Notes

- Use **Long Fast** for maximum text range. PTT voice clips work well on this link.
- Live voice calls require **Short Fast** or faster, which reduces range. Half-duplex mode is recommended.
- The transport node is unattended and holds no messages. Offline devices receive messages when their host reconnects and a fresh path is established.

Scenario 2 — Liberty Chat and LCS MeshChat connected through the internet

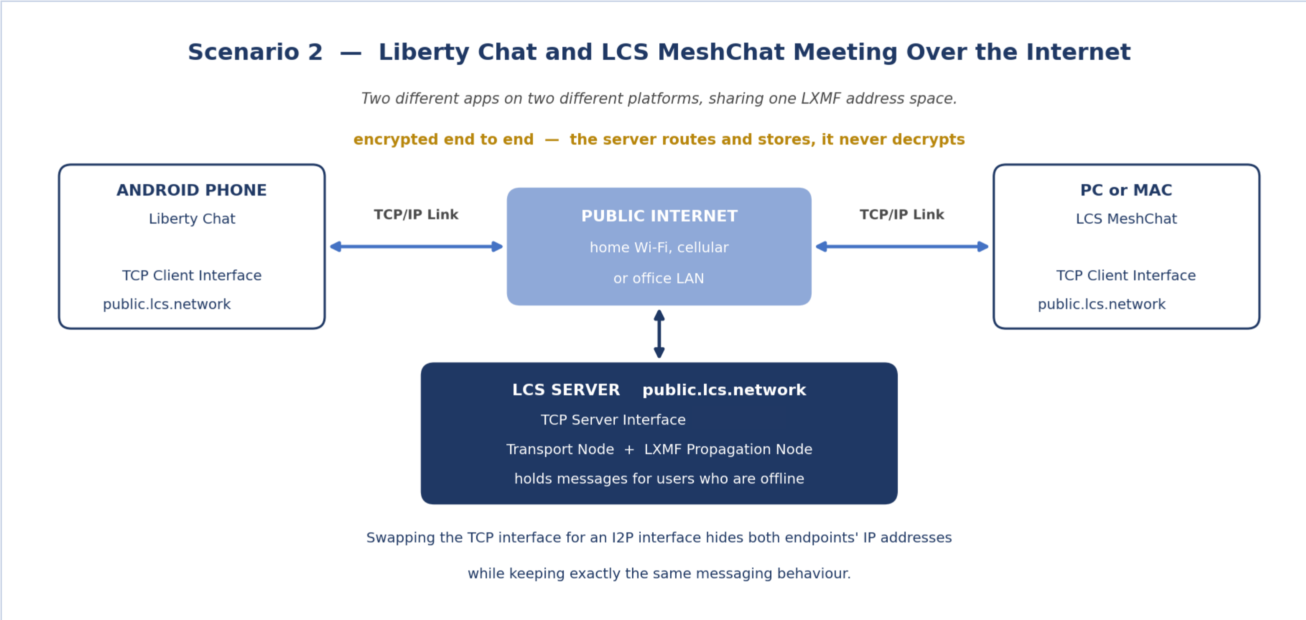

Figure 2 — Two applications meeting on the LCS TCP server.

Requirements

- An Android phone running Liberty Chat, with any internet connection
- A Windows or Mac computer running LCS MeshChat
- A TCP Client interface on each, pointed at public.lcs.network
- No radio hardware

How it works

- Both apps open an outbound TCP connection to the LCS server. No public IP, static address, or port forwarding is needed on either end.
- The server acts as a transport node, learning both announces and routing between them. As a propagation node, it holds messages for offline users and delivers them on reconnection.
- Liberty Chat and LCS MeshChat are different applications on different operating systems, but both speak LXMF. Contacts, delivery receipts, attachments, and voice all function normally between them.

Notes

- This is the fastest way to verify the network before radio hardware is available.
- Bandwidth over TCP supports Opus high-bandwidth voice and full-quality live calls.
- Replacing the TCP interface with I2P hides both endpoints' IP addresses. Application behaviour is unchanged.

Scenario 3 — Two mobile Command Center Gateways over packet radio

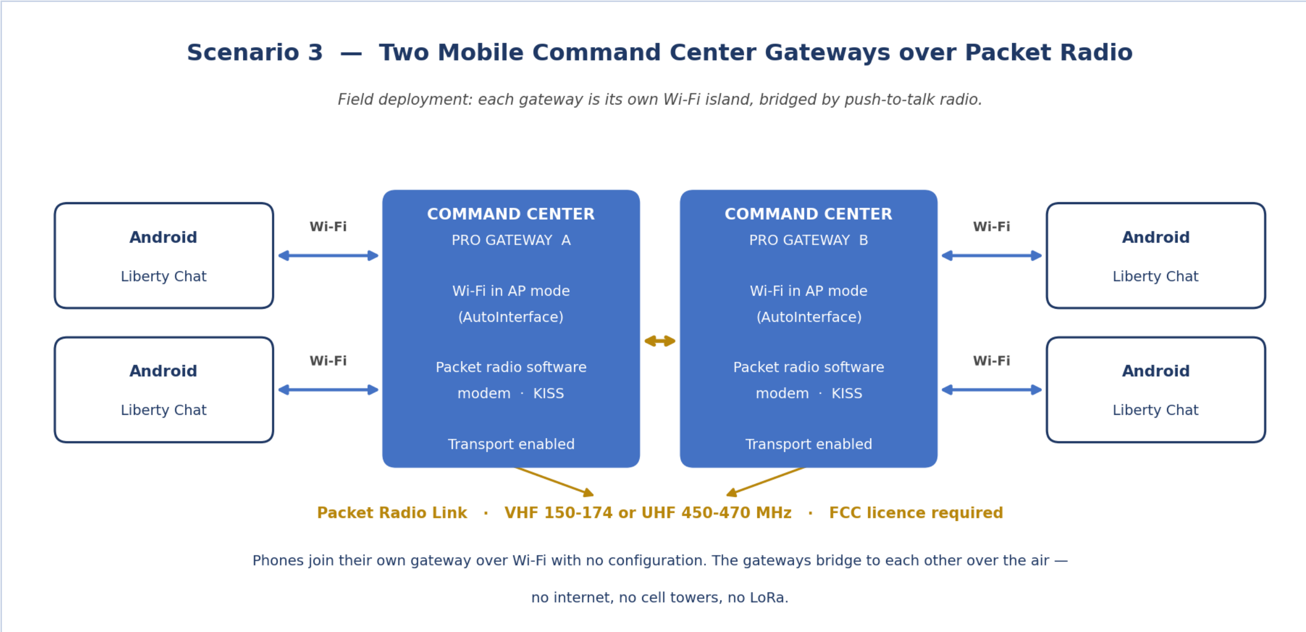

Figure 3 — Field deployment: two Wi-Fi islands bridged by packet radio.

Requirements

- Two Command Center PRO Gateways in Wi-Fi AP mode
- A push-to-talk radio on each gateway, driven by the built-in software modem
- Android phones running Liberty Chat at each site, joined to their local gateway's Wi-Fi
- FCC licence for encrypted operation on 150–174 MHz or 450–470 MHz

How it works

- Each gateway broadcasts its own Wi-Fi network. Phones join via AutoInterface with no IP configuration required.
- The gateways link to each other over packet radio using KISS framing through the software modem.
- Announces from Site A propagate to Site B over the RF link; messages route automatically in both directions.

Notes

- Packet radio provides greater range than LoRa at comparable power, making it the appropriate long-haul medium here.
- The packet radio link is half-duplex and slower than LoRa at short range — expect text and PTT voice clips across the RF hop rather than live calls.
- Each gateway holds mail for its own site. A phone that loses power receives its queued messages when it rejoins the Wi-Fi network.
- Assign a shared network name and passphrase to close the deployment to devices on the same channel.

Scenario 4 — Two homes with IP RNodes on the local Wi-Fi

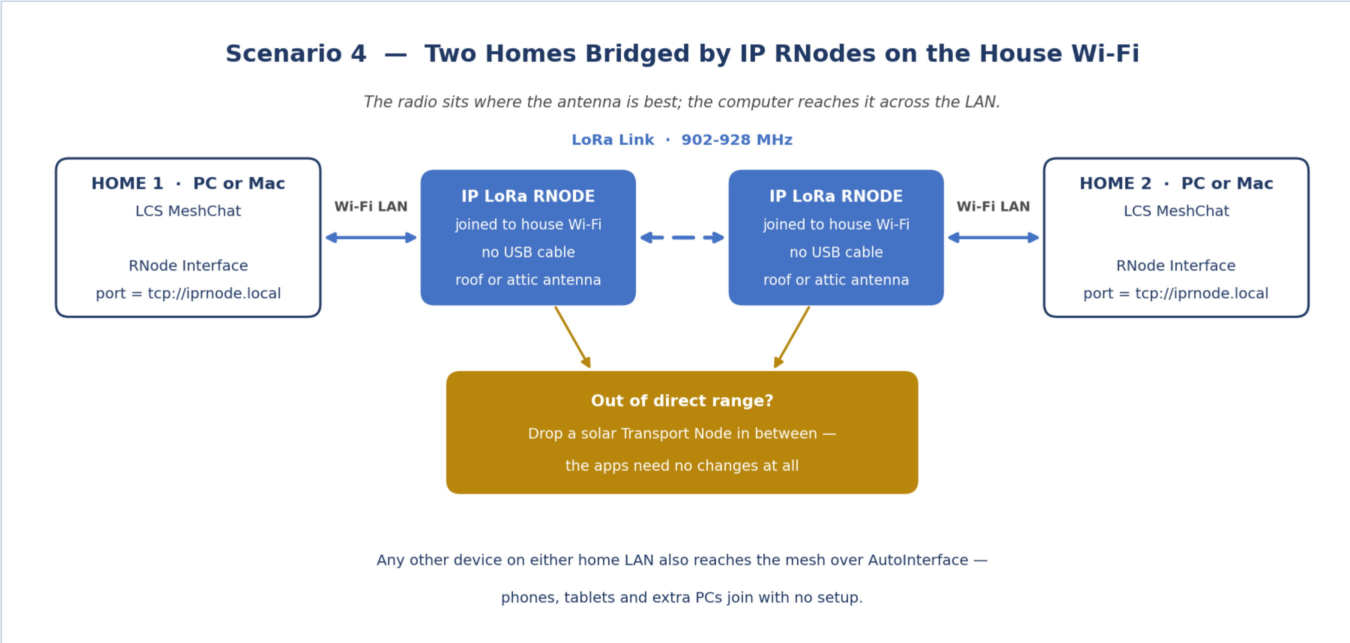

Figure 4 — The radio near the antenna, the application anywhere on the LAN.

Requirements

- One IP LoRa RNode per house, joined to that house's Wi-Fi network
- LCS MeshChat on a PC or Mac in each house
- An RNode interface in MeshChat configured with port = tcp://iprnode.local

How it works

- The RNode is accessed over the LAN, allowing it to be positioned in an attic or near a window for antenna performance while the computer stays at a desk.
- LCS MeshChat treats the Wi-Fi-connected RNode identically to a USB-connected unit.
- The two RNodes communicate directly over LoRa, carrying LXMF traffic between houses.

Notes

- Any other device on either home LAN — phones, tablets, additional computers — can also participate via AutoInterface with no configuration.
- If the two homes are out of direct LoRa range, add a solar transport node between them. No application changes are required; the path simply becomes two hops.
- Run LCS MeshChat as a propagation node on at least one machine so that offline devices receive queued messages on reconnection.

Scenario 5 — Two RNode user clusters joined by packet radio

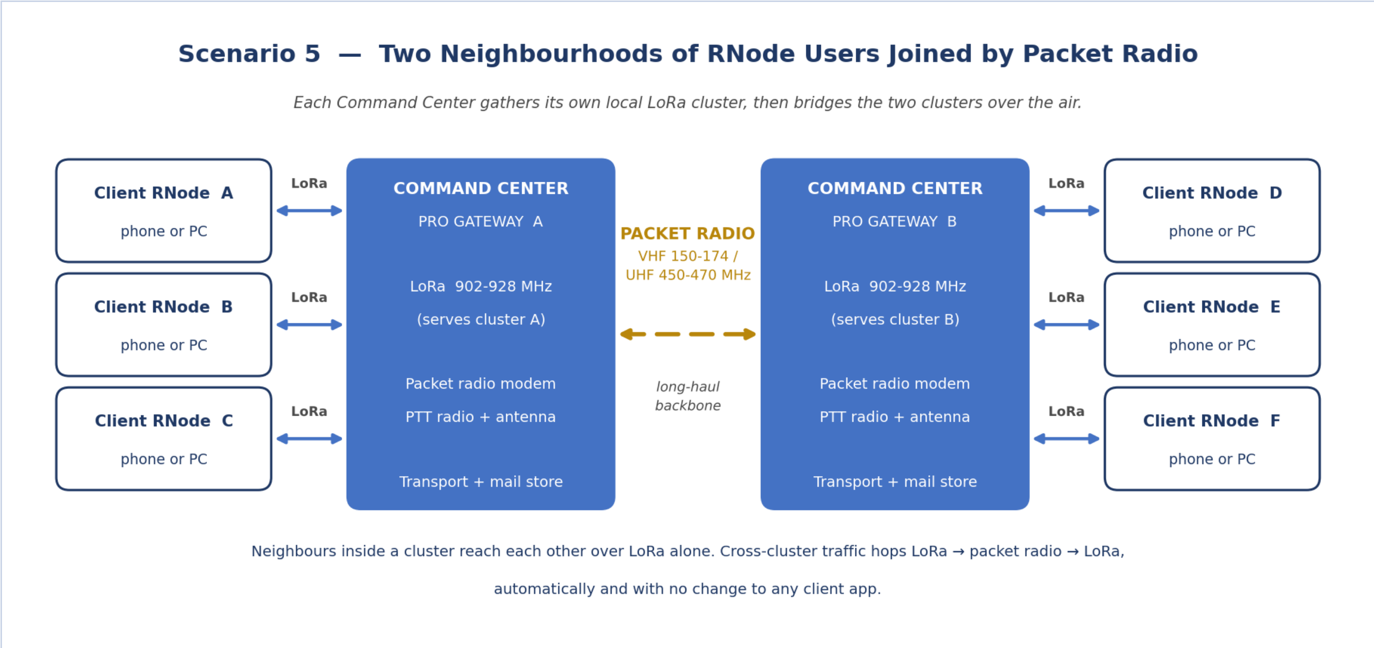

Figure 5 — Two local LoRa clusters bridged over packet radio.

This topology connects two geographically separate groups — neighbouring towns, two ends of a valley, a base camp and a forward site. Each Command Center Gateway serves a local cluster of RNode users over LoRa. The packet radio link carries cross-cluster traffic between the two gateways.

Requirements

- Two Command Center PRO Gateways, positioned for best antenna performance at each site
- A push-to-talk radio on each gateway, driven by the built-in software modem
- A cluster of LCS Client RNodes at each site — one per user, connected by USB-C or Bluetooth to a phone or computer
- FCC licence for encrypted operation on the business band

How it works

- **Within a cluster,** users reach the gateway and often each other directly over LoRa. Local traffic does not touch the packet radio link.
- **Between clusters,** each gateway bridges LoRa to packet radio. A message from Site A to Site D crosses three hops: LoRa in, packet radio across, LoRa out.
- Announces propagate the same path. Users at Site A see Site B users appear in their peer list by name without any manual routing step.
- Both gateways run as transport nodes and propagation nodes, holding mail for their own segment independently.

Why this topology

- **Each radio medium is used where it is most effective.** LoRa is licence-free and suited to short local links. Packet radio provides substantially greater range on comparable power for the single long hop between sites.
- **The backbone stays quiet.** Most traffic is local and remains on LoRa. The packet radio link carries only cross-cluster traffic.

- **It scales by adding gateways.** A third site joins by placing a third Command Center on the same packet radio channel; no client device reconfiguration is required.
- **Individual segments remain operational if the backbone link fails.** Cross-site messages queue at each propagation node and are delivered when the link is restored.

Notes

- All RNodes within a cluster must share the same frequency, bandwidth, spreading factor, and coding rate. A mismatch produces silence rather than a degraded link.
- Live voice calls between sites are marginal over the packet radio hop. Within a cluster on a fast LoRa preset they are more feasible.

13. Requirements Checklist

Every device

- An **identity** — generated at first launch, backed up to a safe location
- At least one **interface** — a radio, a Wi-Fi connection, or a TCP server address
- An **announce** — sent once at setup, and re-sent after any change in connection method

For offline delivery

- At least one **propagation node** that remains powered — a Command Center Gateway, the LCS server, or a desktop machine running LCS MeshChat

For extended range

- One or more **transport nodes** with enable_transport = Yes, positioned high
- Matching radio parameters on every device sharing a channel — frequency, bandwidth, spreading factor, coding rate. A mismatch produces no connection, not a slow one.

For internet access

- A TCP or I2P interface on at least one device with internet access. All devices behind it can route through it.

For live voice

- A link with sufficient capacity: Wi-Fi, Ethernet, TCP, or LoRa at Short Fast or faster. Otherwise use PTT voice clips.

Legal

- LoRa at 902–928 MHz is licence-free in the United States.
- Encrypted operation on VHF/UHF business-band frequencies requires an FCC licence. Regulations vary by country.

Not required

- Accounts, subscriptions, SIM cards, or phone numbers
- DNS, DHCP, port forwarding, or static IP addresses
- Permission from any central authority to join, extend, or operate the network

Reference Links

- Liberty Chat (for Android) — https://github.com/daylight-hub/libertychat

- LCS MeshChat (Windows, Linux, & MacOS) — https://github.com/daylight-hub/reticulum-meshchat
- Liberty Communication Systems, Inc. — https://lcs.network/

⚠ RF RADIATION WARNING (WI-FI / LORA RADIO)

RF RADIATION WARNING: This product transmits radiofrequency (RF) electromagnetic radiation via Wi-Fi and/or LoRa radio protocols. The International Agency for Research on Cancer (IARC) has classified RF radiation as a possible human carcinogen (Group 2B). Peer-reviewed research has identified potential associations between RF electromagnetic field (RF-EMF) exposure and adverse effects on human fertility, including possible reductions in sperm count and pregnancy rates. Individuals with pacemakers, implantable cardiac defibrillators, or other active medical implants, or those with known heart conditions, should consult a physician before use, as RF radiation may interfere with implanted devices. Prolonged close-range exposure should be minimized as a precautionary measure. LIBERTY COMMUNICATION SYSTEMS, INC. EXPRESSLY DISCLAIM ALL LIABILITY FOR ANY HEALTH EFFECTS, INJURY, OR DAMAGE ARISING FROM RF EXPOSURE ASSOCIATED WITH THIS PRODUCT. By using this product, the user assumes all risk associated with RF exposure and acknowledges having read this warning.

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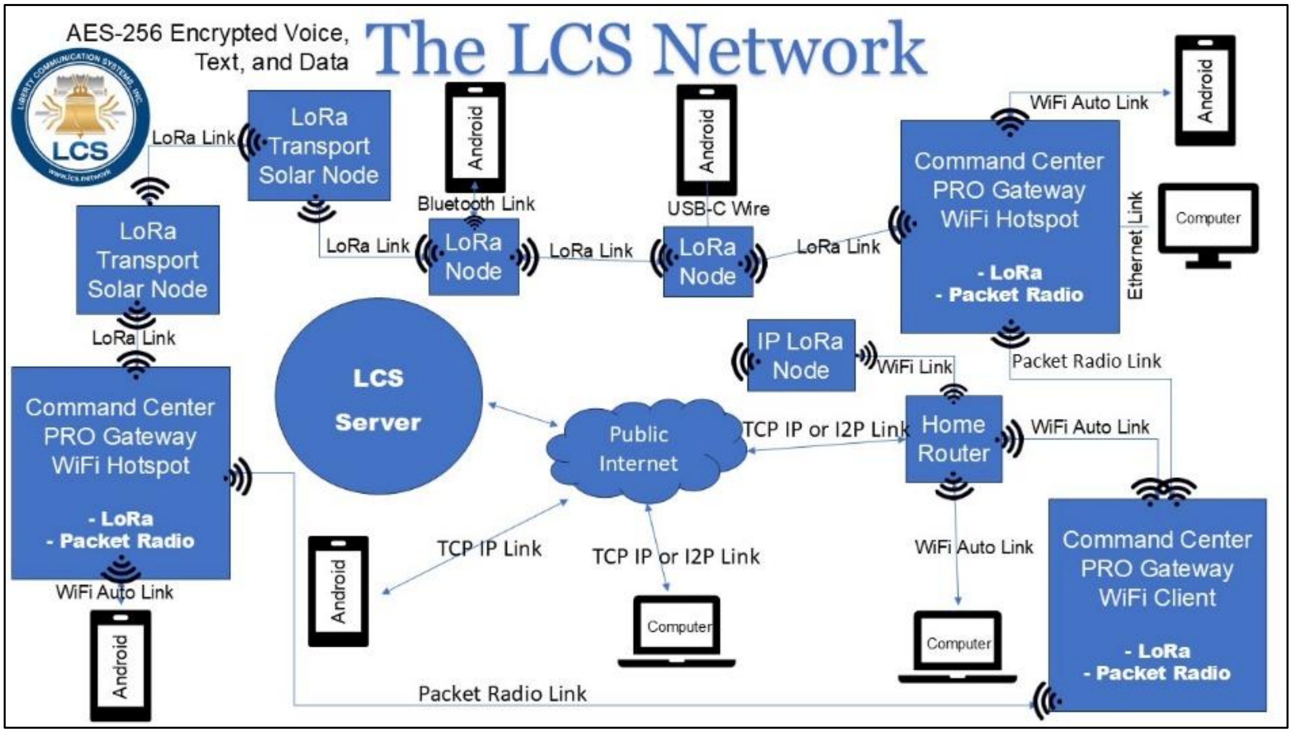
AES-256 Encrypted Voice, Text, and Data
The LCS Network
LCS
LoRa Link
LoRa Transport Solar Node
Android
Bluetooth Link
LoRa Link
LoRa Node
LoRa Link
USB-C Wire
LoRa Node
LoRa Link
WiFi Auto Link
Command Center PRO Gateway WiFi Hotspot
- LoRa
- Packet Radio
Ethernet Link
Computer
LoRa Transport Solar Node
LoRa Link
IP LoRa Node
WiFi Link
Packet Radio Link
Command Center PRO Gateway WiFi Hotspot
- LoRa
- Packet Radio
LCS Server
Public Internet
TCP IP or I2P Link
Home Router
WiFi Auto Link
TCP IP Link
TCP IP or I2P Link
WiFi Auto Link
Command Center PRO Gateway WiFi Client
- LoRa
- Packet Radio
WiFi Auto Link
Computer
Packet Radio Link